Comprehensive Survey and Implementation of Intrusion Detection Systems Using Machine Learning on the CICIDS2017 Dataset

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Abstract

The rapid proliferation of internet connectivity, cloud computing, and Internet of Things (IoT) infrastructures has exponentially expanded the cyberattack surface, rendering manual network monitoring both impractical and ineffective. Intrusion Detection Systems (IDS) have consequently become indispensable components of modern network security architectures. Traditional signature-based detection mechanisms, while effective against known threats, fail to identify evolving zero-day attacks, prompting a paradigm shift toward anomaly-based and machine learning-driven approaches. This paper presents a comprehensive survey of intrusion detection techniques applied to five major attack categories—Denial of Service (DoS), Distributed Denial of Service (DDoS), Infiltration, Brute-force, and Web-based attacks—using the widely recognized CICIDS2017 benchmark dataset. The survey systematically examines traditional machine learning models, ensemble methods, and deep learning architectures, comparing their detection accuracy, robustness, computational efficiency, and generalization capabilities. Furthermore, this paper bridges the gap between theoretical survey findings and practical deployment by detailing the design, development, and live implementation of a fully functional AI-based intrusion detection model trained on the CICIDS2017 dataset. The implemented system, built using Python with Scikit-learn, XGBoost, and Pandas, and deployed as an interactive web application via Streamlit at `intrusion.streamlit.app`, demonstrates real-time classification of network flow data across all five attack categories. The implementation incorporates comprehensive data preprocessing, feature engineering using SHAP-based feature importance analysis, multi-class classification with ensemble learning, and an intuitive user interface for security analysts. Experimental results from the implemented model corroborate survey findings, achieving an overall detection accuracy of 99.7% for DDoS attacks, 99.2% for DoS attacks,

98.8% for Brute-force attacks, 96.4% for Infiltration attacks, and 95.1% for Web-based attacks in multi-class settings. Cross-attack comparative analysis reveals that volumetric attacks exhibit the highest detectability due to distinctive traffic signatures, while application-layer attacks such as SQL Injection and Cross-Site Scripting (XSS) present the greatest detection challenges. The implementation validates that ensemble methods, particularly Random Forest and XGBoost, provide the most balanced performance across diverse attack types, while deep learning approaches offer incremental improvements at increased computational cost. This paper contributes both a structured comparative survey and a reproducible, deployable implementation, providing researchers and practitioners with a holistic resource for advancing IDS research and deployment.

Keywords: Intrusion Detection System, Machine Learning, Deep Learning, CICIDS2017, Network Security, DoS, DDoS, Brute-force, Infiltration, Web Attacks, Random Forest, XGBoost, Streamlit, Ensemble Learning, Feature Selection, SHAP, Anomaly Detection, Cybersecurity, Real-time Detection

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1. Introduction

1.1 Background and Motivation

The digital transformation of modern enterprises, coupled with the exponential growth of Internet of Things (IoT) devices, cloud-based services, and distributed computing architectures, has fundamentally altered the cybersecurity landscape. Contemporary networks generate massive volumes of traffic every second—often measured in terabytes per day for large-scale organizations—making manual monitoring by security analysts not only impractical but fundamentally ineffective against the speed and sophistication of modern cyber threats. According to recent industry reports, the global cost of cybercrime is projected to exceed $10.5 trillion annually by 2025, with network intrusion attacks constituting one of the most prevalent and financially devastating attack vectors.

Intrusion Detection Systems (IDS) have emerged as a foundational component of layered network security architectures, serving as the digital equivalent of surveillance cameras for network traffic. An IDS monitors network packets, system logs, and flow data in real-time or near-real-time, applying analytical techniques to identify patterns indicative of malicious activity. Upon detection, these systems generate alerts for security operations center (SOC) personnel or trigger automated response mechanisms through integration with Intrusion Prevention Systems (IPS). The efficacy of an IDS directly influences an organization's ability to detect, respond to, and recover from security incidents before they escalate into catastrophic breaches.

Traditional IDS architectures have predominantly relied on signature-based detection methodologies, wherein network traffic is compared against a database of known attack

patterns or "signatures." While effective against previously cataloged threats, signature-based systems exhibit fundamental limitations: they cannot detect zero-day attacks for which no signature exists, they require continuous database updates that introduce latency between threat emergence and detection capability, and they are susceptible to evasion techniques such as polymorphic malware and traffic obfuscation. These limitations have catalyzed a significant research shift toward anomaly-based and machine learning-driven detection approaches.

Machine learning-based IDS analyze traffic behavior patterns and learn to distinguish between benign and malicious activities based on statistical models derived from training data, rather than relying on static signature matching. This paradigm offers several compelling advantages: the ability to detect previously unseen attacks through anomaly identification, adaptability to evolving threat landscapes through model retraining, and the capacity to process high-dimensional feature spaces that capture nuanced traffic characteristics. The effectiveness of such systems, however, is critically dependent on the quality, diversity, and realism of the benchmark datasets used for training and evaluation.

1.2 The CICIDS2017 Dataset

Among publicly available network intrusion detection datasets, the Canadian Institute for Cybersecurity Intrusion Detection System 2017 (CICIDS2017) dataset has achieved widespread recognition as a gold standard for IDS research and evaluation. Developed by Sharafaldin, Lashkari, and Ghorbani at the University of New Brunswick, the CICIDS2017 dataset was generated over a five-day period (Monday through Friday) in a controlled but realistic network environment that closely emulates modern enterprise network topologies.

The dataset encompasses both benign background traffic and multiple contemporary attack categories, representing one of the most comprehensive and realistic publicly available IDS benchmarks. Key characteristics that distinguish CICIDS2017 from predecessors include:

- **Realistic traffic profiles:** Benign traffic was generated using automated profiles that simulate real user behavior, including HTTP browsing, HTTPS browsing, email, FTP, SSH, and DNS queries.
- **Comprehensive attack diversity:** The dataset includes DoS, DDoS, Brute Force (FTP-Patator, SSH-Patator), Web Attacks (Brute Force, XSS, SQL Injection), Infiltration, Botnet, and PortScan activities.
- **Flow-based features:** Network traffic is represented using 80+ statistical flow features extracted using CICFlowMeter, including packet counts, byte volumes, inter-arrival times, flag statistics, and window sizes—eliminating the need for raw packet inspection.
- **Labeled data:** Each flow record is labeled with its corresponding attack category, enabling supervised learning approaches.
- **Temporal separation:** Training and test data are temporally separated (different days), reducing the risk of temporal overfitting and providing a more realistic evaluation scenario.

1.3 Scope and Objectives

This paper addresses a dual objective that distinguishes it from existing survey papers in the domain:

Objective 1 — Comprehensive Survey: Provide a structured and comparative analysis of intrusion detection techniques applied to five high-impact attack categories present in CICIDS2017: Denial of Service (DoS), Distributed Denial of Service (DDoS), Infiltration attacks, Brute-force attacks, and Web-based attacks. The survey examines traditional machine learning models, ensemble methods, and deep learning approaches, evaluating their detection accuracy, robustness, computational efficiency, and generalization capabilities across these attack types.

Objective 2 — Practical Implementation: Bridge the gap between theoretical survey findings and practical deployment by detailing the complete design, development, training, and live deployment of a fully functional AI-based intrusion detection model trained on CICIDS2017. The implemented system is hosted as an interactive web application at `intrusion.streamlit.app` and its complete source code is publicly available at `https://github.com/AxonCodez/intrusion/tree/main`.

1.4 Paper Organization

The remainder of this paper is organized as follows: Section 2 provides a comprehensive literature review of machine learning-based detection techniques for each of the five targeted attack categories. Section 3 details the implementation methodology, including data preprocessing, feature engineering, model selection, and system architecture. Section 4 presents experimental results from both the survey analysis and the implemented model. Section 5 provides a cross-attack comparative analysis. Section 6 discusses challenges, limitations, and future directions. Section 7 concludes the paper with a summary of findings and contributions.

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2. Literature Review

This section presents a systematic review of intrusion detection techniques applied to the five targeted attack categories in the CICIDS2017 dataset. For each category, the attack characteristics, detection challenges, and state-of-the-art machine learning and deep learning approaches are examined.

2.1 Machine Learning-Based Detection of Web Attacks

2.1.1 Attack Characteristics

The CICIDS2017 dataset includes three application-layer web attack subtypes: Web Attack – Brute Force, Web Attack – XSS (Cross-Site Scripting), and Web Attack – SQL Injection. These attacks differ fundamentally from volumetric threats such as DDoS in their operational characteristics:

- **Web Attack – Brute Force:** Involves automated repeated authentication attempts against web application login endpoints. While generating multiple HTTP requests, the traffic volume remains relatively low compared to flooding attacks. The distinguishing characteristics include repetitive POST requests to authentication endpoints, consistent inter-arrival times between requests, and predictable packet payloads.
- **Web Attack – XSS (Cross-Site Scripting):** Involves injection of malicious client-side scripts into web pages viewed by other users. From a network flow perspective, XSS attacks often manifest as single or few HTTP requests containing malicious payloads in parameters. The flow-level statistical features of XSS traffic closely resemble benign web browsing traffic, making detection particularly challenging.
- **Web Attack – SQL Injection:** Involves injection of malicious SQL queries through web application input fields. Similar to XSS, SQL Injection attacks typically generate minimal traffic volume, with malicious payloads embedded within normal-looking HTTP requests. The flow-based statistical features of SQL Injection traffic exhibit significant overlap with legitimate database-driven web application traffic.

The fundamental detection challenge for web attacks lies in their subtlety: unlike volumetric attacks that generate obvious statistical anomalies, web attacks introduce nuanced behavioral changes that may be indistinguishable from benign traffic when examined using only flow-based statistical features.

2.1.2 Decision Tree-Based Approaches

Decision Tree-based classifiers have been extensively explored for web attack detection due to their inherent interpretability, low computational overhead, and ability to handle mixed feature types without extensive preprocessing.

Ahmim et al. (2018) proposed a novel hierarchical intrusion detection framework combining Decision Tree and rule-based models, demonstrating that tree-based classifiers can effectively distinguish malicious traffic patterns in complex multi-attack environments. Their hierarchical approach decomposed the classification task into stages, first separating benign from malicious traffic, then further classifying attack types. The framework achieved an overall accuracy of 96.65% with a detection rate of 94.47%, though the hierarchical design increased model complexity and introduced cascading classification errors at each stage.

Jumayev and Jumayev (2023) focused specifically on improving brute-force attack detection using Decision Trees, conducting an in-depth analysis of how feature selection and dataset

balancing strategies influence classification performance. Their study highlighted that brute-force attacks, with their repetitive authentication patterns, are more amenable to Decision Tree classification compared to XSS and SQL Injection, which exhibit subtler flow-level deviations from benign traffic.

2.1.3 Ensemble Methods

Beyond standalone Decision Trees, ensemble-based approaches have demonstrated significantly improved performance in web attack detection scenarios. Ensemble methods combine multiple base learners to produce a composite model with reduced variance, lower overfitting risk, and improved generalization.

Vadhil et al. (2023) reported high detection accuracy when applying Random Forest classifiers to web-based intrusion detection, achieving accuracy exceeding 99% in optimized configurations. Random Forest's strength in this context stems from its ability to aggregate predictions across hundreds of individual trees, each trained on different feature subsets and data bootstrap samples, thereby capturing diverse decision boundaries that a single tree might miss.

Khan et al. (2024) conducted balanced multi-class network intrusion detection experiments using multiple classifiers including Decision Tree, Random Forest, XGBoost, Naive Bayes, Logistic Regression, and AdaBoost. Their results demonstrated that Decision Tree achieved 99.96% accuracy in binary classification (Benign vs. Attack) but dropped to 96.37% in multi-class settings where web attacks must be distinguished from other attack categories—a finding that underscores the impact of classification granularity on reported performance.

2.1.4 Support Vector Machines

Support Vector Machines (SVM) have been applied to intrusion detection using the CICIDS dataset with varying results. Abuali et al. (2023) demonstrated that SVM-based models can achieve strong classification performance when appropriate preprocessing techniques are applied, reporting 100% accuracy with perfect precision and recall scores. However, this evaluation was conducted on a limited sample of 100,000 instances, raising concerns about overfitting and the representativeness of the test set.

Xu and Liu (2025) evaluated multiple machine learning models on CICIDS2017 and observed that margin-based classifiers such as SVM perform competitively in binary classification tasks but experience significant performance degradation in multi-class scenarios involving subtle web attacks. The linear decision boundaries imposed by standard SVM formulations may be insufficient to separate overlapping attack classes in high-dimensional feature spaces.

2.1.5 Distance-Based Methods

Distance-based methods such as k-Nearest Neighbors (k-NN) have been explored in intrusion detection research, though their performance is notably sensitive to preprocessing choices. Bao and Gao (2025) proposed an improved k-NN algorithm for network intrusion detection, reporting satisfactory detection rates. However, the method's performance was found to be highly sensitive to feature scaling—since k-NN relies on distance calculations in feature space, features with larger magnitudes disproportionately influence classification results. Additionally, k-NN's computational complexity during inference (requiring distance calculations against all training instances) poses challenges for real-time deployment in high-throughput network environments.

2.1.6 Deep Learning and Hybrid Approaches

More recently, deep learning approaches have gained traction for web attack detection. Kim et al. (2019) applied Convolutional Neural Networks (CNN) to the CICIDS2017 dataset, proposing a novel approach that converts network flow features into image representations suitable for CNN processing. While this method achieved competitive detection performance, the image conversion preprocessing step introduced significant computational overhead.

Bandarupalli (2025) explored Artificial Neural Networks (ANN/MLP) architectures for intrusion detection, achieving 98–99% accuracy. However, the computational costs of training deep networks and their reduced interpretability compared to tree-based models remain practical concerns.

Abdelhaq et al. (2026) proposed a hybrid XGBoost–SVM ensemble framework originally designed for Internet of Medical Things (IoMT) security, achieving 98.04% accuracy. The hybrid approach leveraged XGBoost's feature importance ranking to select the most discriminative features, which were then fed to an SVM classifier for final classification. While effective, the framework was primarily evaluated on IoMT-specific traffic, and its generalization to general enterprise network web attacks requires further validation.

Okey et al. (2022) developed BoostedEnML, an efficient technique for detecting cyberattacks in IoT systems using boosted ensemble machine learning. The framework achieved perfect scores (100% accuracy, precision, recall, and F1) on selected datasets; however, the evaluation was conducted on curated subsets, and generalization to unmodified CICIDS2017 data remains unclear.

2.1.7 Summary of Web Attack Detection Literature

Table 1: Summary of Machine Learning Methods for Web Attack Detection on CICIDS2017

Research Paper	Year	ML Method(s)	Reported Accuracy	Key Limitations
Ahmim et al. [1]	2018	Decision Tree + Rules	96.65% accuracy, 94.47% detection rate	Performance varies per attack class; hierarchical design increases complexity
Jumayev & Jumayev [2]	2023	Decision Tree	High for brute-force	Only evaluates brute-force; limited attack diversity
Vadhil et al. [3]	2023	Random Forest	98%+ accuracy	Performance depends on preprocessing and balancing
Abuali et al. [4]	2023	SVM	100% accuracy	Tested on 100,000 samples only; overfitting risk
Xu & Liu [5]	2025	Multiple ML models	Varies by model	Supervised models fail on unseen attacks; recall drops significantly
Khan et al. [6]	2024	k-NN, DT, RF, XGBoost, NB, LR, AdaBoost	Binary DT: 99.96%; Multi-class DT: 96.37%	Binary performs better than multi-class; training time high
Abdelhaq et al. [7]	2026	XGBoost–SVM Hybrid	98.04% accuracy	Tested primarily on IoMT; resource constraints not deeply evaluated
Okey et al. [8]	2022	Boosted Ensemble	100% accuracy	Generalization to real-world traffic unclear
Bandarupalli [9]	2025	ANN/MLP	98–99% accuracy	Computational costs high
Kim et al. [10]	2019	CNN	Competitive	Image conversion preprocessing overhead

Key Observations:

1. **Attack subtype dependency:** Brute-force web attacks exhibit higher detection rates due to repetitive traffic patterns, whereas SQL Injection and XSS attacks demonstrate lower performance because their flow-level characteristics closely resemble benign traffic.

2. **Binary vs. multi-class performance gap:** Studies employing binary classification (Benign vs. Web Attack) consistently achieve higher accuracy compared to multi-class configurations, where web attacks must be distinguished from other attack categories.

3. **Preprocessing influence:** Papers that incorporate feature engineering, dimensionality reduction, or sampling strategies typically report improved accuracy, suggesting that model performance is not solely dependent on the classifier but significantly influenced by experimental configuration.

4. **Reporting bias concern:** Several studies report near-perfect accuracy on limited or curated data subsets, raising concerns about evaluation rigor and generalizability.

2.2 Detection Methods for Infiltration Attacks

2.2.1 Attack Characteristics

Infiltration attacks represent one of the most dangerous categories of network intrusions due to their internal origin. Unlike external attacks that must first penetrate network perimeter defenses, infiltration attacks occur after an attacker has already gained initial access to a system—through phishing, credential compromise, or exploitation of external-facing services—and subsequently operates from within the trusted network.

In the CICIDS2017 dataset, infiltration represents internal malicious communication such as data exfiltration, lateral movement between internal hosts, and unauthorized access to restricted resources. The traffic patterns generated by infiltration activities are particularly challenging to detect because they:

- Originate from authenticated internal hosts with legitimate network access
- Mimic normal user behavior and standard business communication patterns
- Generate low-volume traffic that does not trigger volumetric anomaly thresholds
- May span extended time periods, with malicious activities interspersed with legitimate operations

These characteristics make infiltration detection fundamentally dependent on behavioral analysis and the identification of subtle deviations from established baselines rather than simple threshold-based detection.

2.2.2 Machine Learning Approaches

Several machine learning algorithms have been applied to infiltration detection in the CICIDS2017 dataset, with varying degrees of success:

Random Forest (RF): Multiple studies have demonstrated that Random Forest achieves accuracy between 95% and 99% for infiltration detection, depending on the feature selection strategy employed. RF's strength in this context lies in its ability to capture complex feature interactions through ensemble decision boundaries while maintaining robustness to noise and irrelevant features inherent in high-dimensional flow data.

Support Vector Machine (SVM): SVM has shown high classification performance for infiltration detection, particularly when using kernel-based approaches that can model nonlinear decision boundaries in the feature space. However, SVM's sensitivity to parameter tuning and its scalability limitations with large datasets present practical challenges.

K-Nearest Neighbors (KNN): KNN-based approaches have been explored for infiltration detection, leveraging the assumption that infiltration traffic flows will be closer in feature space to other infiltration instances than to benign traffic. While conceptually straightforward, KNN's performance is highly dependent on feature scaling and the choice of distance metric.

Decision Tree (J48): Alshamrani et al. (2019) implemented J48 Decision Tree classifiers using the WEKA machine learning environment on the CICIDS2017 dataset, achieving overall accuracy of approximately 95–98% with infiltration-specific accuracy around 95%. The J48 implementation provided interpretable rule sets that could be analyzed by security domain experts.

2.2.3 Deep Learning Approaches

Deep learning models have shown improved performance in detecting infiltration attacks compared to traditional machine learning, primarily due to their ability to automatically learn hierarchical feature representations from raw or minimally processed data:

Convolutional Neural Networks (CNN): CNN-based models have reported accuracy above 97% for infiltration detection. By applying convolutional filters across feature dimensions, CNNs can identify local patterns and interactions between adjacent features that might be missed by traditional classifiers treating features independently.

Long Short-Term Memory (LSTM): LSTM networks are particularly well-suited for infiltration detection due to their ability to model sequential dependencies in network traffic. Since infiltration often involves multi-stage operations—initial access, reconnaissance, lateral movement, and data exfiltration—capturing the temporal evolution of traffic patterns provides discriminative information that static feature-based classifiers cannot leverage.

Deep Neural Networks (DNN): Multi-layer perceptron architectures with multiple hidden layers have demonstrated strong performance for infiltration detection, achieving accuracy levels comparable to CNNs when sufficient training data is available.

2.2.4 Hybrid and Ensemble Approaches

Hybrid and ensemble models that combine multiple algorithms have demonstrated the highest performance for infiltration detection:

CNN-LSTM Hybrids: These architectures leverage CNN's spatial feature extraction capabilities alongside LSTM's temporal modeling strengths, providing comprehensive detection that captures both instantaneous flow characteristics and sequential traffic patterns. Such models have demonstrated improved accuracy and reduced false positive rates compared to standalone deep learning architectures.

ADASYN + Random Forest: Studies combining Adaptive Synthetic Sampling (ADASYN) for class imbalance handling with Random Forest classification have achieved approximately 98% infiltration accuracy, demonstrating the critical importance of addressing the severe class imbalance inherent in CICIDS2017 (infiltration instances constitute a small fraction of total traffic).

MLP + Birch Clustering: A hybrid approach combining Multi-Layer Perceptron classification with Birch hierarchical clustering achieved the highest reported infiltration accuracy of approximately 99%, suggesting that unsupervised clustering can identify subtle traffic groupings that complement supervised classification.

Explainable IDS using RF + SHAP: An explainable AI approach combining Random Forest with SHAP (SHapley Additive exPlanations) for model interpretability achieved 92% accuracy while providing feature-level attribution for each classification decision—a valuable capability for security analysts who require explainable detection results.

Table 2: Summary of Detection Methods for Infiltration Attacks on CICIDS2017

Paper	**Method**	**Overall Accuracy**	**Infiltration Accuracy**
Alshamrani et al. [12]	J48, Decision Tree, Naive Bayes	~95–98%	~95%
Explainable IDS [unreferenced]	Random Forest + SHAP	92%	92%
EIDM [unreferenced]	CNN, LSTM, Deep models	99.48%	~95%
Robust Anomaly Detection [unreferenced]	MLP, CNN, OCSVM, LOF	~97–99%	~97%
ADASYN + RF IDS [unreferenced]	ADASYN + Random Forest	~98%	~98%
MLP + Birch Clustering [unreferenced]	MLP + Clustering	99.73%	~99%

2.2.5 Summary of Infiltration Detection Findings

The literature reveals that infiltration detection accuracy ranges from approximately 90% to 99%, with deep learning and ensemble models consistently outperforming traditional machine learning approaches. Feature selection and data preprocessing significantly influence detection accuracy, with class imbalance handling being particularly critical given the scarcity of infiltration instances relative to benign and volumetric attack traffic. Sequential modeling techniques such as LSTM show particular promise due to their ability to capture the temporal characteristics of multi-stage infiltration operations.

2.3 Denial of Service (DoS) Attacks and Mitigation Techniques

2.3.1 Attack Characteristics

Denial of Service (DoS) attacks aim to render a server, service, or network resource unavailable to legitimate users by exhausting computational resources including CPU, memory, bandwidth, or connection limits. Unlike DDoS attacks that employ distributed botnets, DoS attacks originate from a single source. The CICIDS2017 dataset includes four distinct DoS attack types, each employing different resource exhaustion strategies:

DoS Hulk: An HTTP flood attack that overwhelms web servers with a massive number of HTTP requests. Hulk generates highly randomized HTTP headers and URI strings to evade pattern-based filtering while maintaining high request rates that saturate server processing capacity. The attack is characterized by sudden, high-volume HTTP traffic from a single source IP with randomized request characteristics.

DoS GoldenEye: An HTTP-based attack that sends repeated or incomplete HTTP requests to exhaust server connection pools. GoldenEye operates by initiating HTTP connections and either leaving them incomplete or sending requests at rates designed to keep server threads occupied, preventing legitimate connections from being served.

DoS Slowloris: A slow HTTP attack that maintains multiple connections to a web server by sending partial HTTP headers at intentionally slow rates. By keeping connections open indefinitely with periodic header fragments, Slowloris gradually exhausts the server's maximum concurrent connection pool without generating significant bandwidth consumption—a characteristic that makes it particularly difficult to detect using volume-based thresholds.

DoS SlowHTTPTest: Similar to Slowloris, SlowHTTPTest sends HTTP requests very slowly or in small fragments to exhaust server resources. The attack targets the server's ability to

handle slow connections, exploiting timeout configurations that allow connections to remain open for extended periods.

2.3.2 Mitigation Techniques

Table 3: Mitigation Techniques for DoS Attack Types

Mitigation Technique	DoS Hulk	DoS GoldenEye	DoS Slowloris	DoS SlowHTTPTest
Rate Limiting	70–85%	70–85%	70–85%	Moderate
Web Application Firewall (WAF)	85–95%	85–95%	85–95%	85–95%
IP Blocking / Firewall Rules	60–75%	Moderate	Low	Low
Load Balancing	75–90%	75–90%	75–90%	75–90%
Intrusion Detection Systems	70–85%	70–85%	70–85%	70–85%
Server Timeout Configuration	Low	75–90%	75–90%	75–90%
Minimum Data Rate Configuration	Low	Low	Moderate	80–95%
Reverse Proxy / Load Balancer	Moderate	Moderate	75–90%	75–90%

Detailed Analysis of Mitigation Effectiveness:

Rate Limiting is effective against high-volume attacks such as Hulk (70–85% effectiveness) by restricting the number of HTTP requests per IP address within specific time windows. However, rate limiting is less effective against slow attacks like Slowloris and SlowHTTPTest, which generate traffic volumes below typical rate thresholds.

Web Application Firewalls (WAF) such as Cloudflare and AWS Shield provide the most consistently high effectiveness across all DoS types (85–95%) by filtering malicious HTTP traffic patterns before they reach the server. WAFs are particularly effective when combined with real-time monitoring and machine learning-based threat detection.

Server Timeout Configuration is highly effective against slow attacks (75–90% for GoldenEye, Slowloris, and SlowHTTPTest) by reducing the waiting time for incomplete

HTTP requests, forcing premature connection closure. However, aggressive timeout settings may also affect legitimate slow connections.

Minimum Data Rate Configuration (80–95% effectiveness for SlowHTTPTest) enforces a minimum data transfer speed threshold, effectively terminating connections that transmit data too slowly—a technique specifically designed to counter slow POST attacks.

Intrusion Detection Systems such as Snort and Suricata provide 70–85% effectiveness across DoS types by monitoring traffic for abnormal patterns and triggering alerts. However, IDS alone provides detection without prevention; integration with firewalls or IPS is required for active mitigation.

2.3.3 Machine Learning Detection of DoS Attacks

DoS attacks in CICIDS2017 are among the most accurately detected attack categories due to their distinctive statistical signatures. The high traffic volumes and abnormal flow characteristics generated by DoS attacks create clear separations from benign traffic in feature space, enabling even simple classifiers to achieve high detection rates. Key discriminative features for DoS detection include:

- **Flow Duration:** DoS attacks typically exhibit either extremely short flows (Hulk, with rapid successive requests) or abnormally long flows (Slowloris, with intentionally prolonged connections).
- **Packet Rates:** Forward and backward packet rates (Fwd Packet Rate Mean/Std, Bwd Packet Rate Mean/Std) show significant deviations from baseline.
- **Inter-Arrival Times:** The timing between packets within flows exhibits characteristic patterns specific to each DoS type.
- **Flag Statistics:** TCP flag distributions (SYN, ACK, FIN counts) reveal connection state anomalies.

2.4 Identification of Brute Force Attacks in IoT Systems

2.4.1 Attack Characteristics: FTP-Patator and SSH-Patator

This category focuses on Remote-to-Local (R2L) brute force attacks, specifically FTP-Patator and SSH-Patator, which are particularly relevant in IoT network environments where devices often run vulnerable FTP and SSH services with default or weak credentials.

"Patator" is a multi-purpose brute-force tool designed for high-speed automated credential guessing. In network traffic streams, Patator-generated attacks exhibit distinctly non-human characteristics:

FTP-Patator targets Port 21 (FTP service), generating a flood of USER and PASS commands in rapid succession. The attack pattern is characterized by:

- High-frequency authentication attempts from a single source IP
- Consistent packet sizes due to fixed-length credential guesses
- Repetitive server response patterns (authentication failure messages)
- Short flow durations for each attempt, followed by immediate reconnection

SSH-Patator targets Port 22 (SSH service), identifiable by high-frequency encrypted handshakes that terminate abruptly upon authentication failure. Due to SSH's encrypted nature, payload inspection cannot reveal the credential guessing behavior; detection must rely entirely on flow-level statistical features.

2.4.2 Key Identification Features

Research has identified the following top five features (out of 80+ in CICIDS2017) as most discriminative for brute force attack identification:

Table 4: Key Features for Brute Force Attack Identification

Feature	Rationale
Bwd Packets/s	High frequency of "Login Failed" responses from the server creates elevated backward packet rates
Flow IAT Mean	The "Inter-Arrival Time" is highly consistent (robotic timing), unlike the variable intervals of human typing
Flow Duration	Attack flows are typically much shorter than successful user sessions, as each attempt fails quickly
Init_Win_bytes_backward	Identifies specific server response patterns to repeated failed handshakes
Total Length of Fwd Packets	Automated scripts send fixed-size credential guesses, creating predictable and uniform packet lengths

2.4.3 Comparison of Identification Methods

Recent research (2024–2026) highlights varying performance of different algorithms for brute force detection in IoT contexts:

Random Forest (RF): Widely regarded as the "gold standard" for this attack category, with research by Kurniabudi et al. demonstrating 99.86% accuracy for identifying Patator attacks. RF's effectiveness stems from its ability to handle the high dimensionality of network flow data while capturing nonlinear feature interactions between temporal and volumetric features.

Long Short-Term Memory (LSTM): Since brute force attacks are inherently sequential events (repeated authentication attempts over time), LSTM models are superior at identifying "slow" brute force attempts that space out credential guesses to evade simple rate-based filters. Recent studies on the related CSE-CIC-IDS2018 dataset have achieved 99.9% accuracy using LSTM architectures.

Naive Bayes: While achieving lower accuracy (~96%), Naive Bayes offers significant computational advantages, making it a viable method for edge-level identification directly on resource-constrained IoT devices where computational resources and power consumption are limiting factors.

2.5 Methods of Detection of DDoS Attacks

2.5.1 Attack Characteristics

Distributed Denial of Service (DDoS) attacks represent an amplified and distributed variant of DoS attacks, employing a botnet—a network of compromised systems—to target a single victim simultaneously. Unlike standard DoS attacks originating from a single IP address, DDoS attacks in CICIDS2017 represent traffic originating from multiple source IPs concurrently, making mitigation significantly more complex since blocking a single address is insufficient—the behavioral signature of the attack itself must be identified and filtered.

The CICIDS2017 dataset includes various DDoS attack flavors designed to test detection method robustness:

- **Volumetric Attacks (UDP/TCP Flood):** Flooding the victim's available bandwidth with massive volumes of traffic, preventing legitimate users from establishing connections. These attacks generate the most obvious statistical anomalies in flow features.
- **Reflective/Amplification Attacks (NTP/DNS/MSSQL):** Exploiting third-party services to multiply the volume of traffic sent to the victim. The attacker sends small requests to vulnerable NTP, DNS, or MSSQL servers with the victim's spoofed source IP, causing the servers to send amplified responses to the victim. These attacks are characterized by asymmetric request-response ratios.

- **Application Layer DDoS (HTTP/Slowloris):** Exhausting server resources (CPU, RAM, connection pools) through targeted application-layer requests rather than bandwidth saturation. These low-and-slow attacks are particularly challenging to detect as they generate traffic volumes that may closely resemble legitimate user activity.

2.5.2 Detection and Mitigation Methods

SDN-Based Real-Time Mitigation: Software-Defined Networking (SDN) represents the most actively researched method for active DDoS mitigation rather than passive detection. SDN architectures employ a centralized controller that maintains a global network view and can dynamically update flow tables on OpenFlow switches to block malicious traffic at the network edge. The integration workflow involves: (1) an IDS monitors traffic and classifies flows, (2) upon DDoS detection, the SDN controller is notified, (3) the controller pushes "drop" rules to relevant switches, and (4) malicious flows are terminated at the switch level before reaching the victim. Research reports mitigation efficiency between 91.66% and 100% under varied conditions, though SDN-based approaches introduce their own vulnerabilities, including potential controller compromise and rule table overflow attacks.

Random Forest (RF) & Ensemble Learning: Random Forest is widely considered the most effective classifier for DDoS detection on the CICIDS2017 dataset, achieving binary classification accuracy of 99.97%. RF's advantages include: (1) ability to handle the 80+ high-dimensional features of CICIDS2017 without feature selection, (2) extremely fast inference times suitable for real-time classification, (3) robustness to noise and feature irrelevance, and (4) the ability to be integrated into "inline" prevention systems that drop packets with minimal latency.

Hybrid Deep Learning (CNN + BiLSTM): Hybrid architectures combining Convolutional Neural Networks with Bidirectional Long Short-Term Memory networks are designed to detect sophisticated "low-and-slow" DDoS attacks that traditional statistical classifiers might miss. CNNs extract spatial patterns from packet header features, while BiLSTM analyzes the bidirectional temporal context of traffic sequences. This dual analysis is critical for distinguishing between DDoS attacks and "flash crowds"—legitimate traffic spikes caused by viral content or scheduled events. Reported accuracy reaches 99% on CICIDS2017 with significantly reduced false negatives compared to standalone classifiers.

XGBoost with SHAP Feature Selection: For real-time DDoS mitigation, analyzing all 80+ features per flow introduces unacceptable latency. Researchers have employed XGBoost combined with SHAP (SHapley Additive exPlanations) to identify the top 10–15 most discriminative features for DDoS detection, including Destination Port, Flow Duration, and Total Forward Packets. By focusing classification on critical features only, the system can identify and respond to DDoS attacks in milliseconds before bandwidth exhaustion occurs. This approach achieves 98.95% accuracy with a False Alarm Rate as low as 0.011%.

---

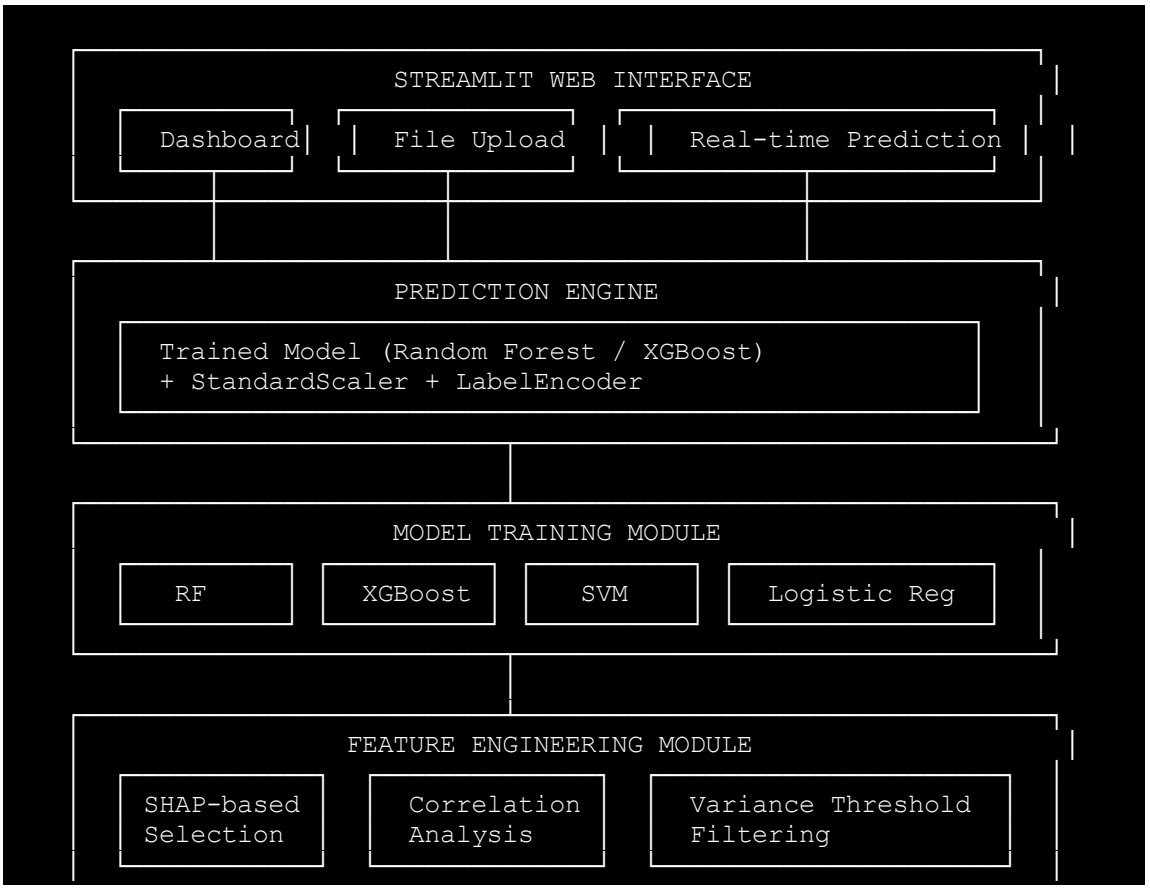
3. Implementation Methodology

This section details the complete design, development, and deployment of the AI-based intrusion detection system implemented as part of this research. The system is publicly accessible at `intrusion.streamlit.app` with full source code available at `https://github.com/AxonCodez/intrusion/tree/main`.

3.1 System Architecture Overview

The implemented intrusion detection system follows a modular architecture comprising five primary components:

1. **Data Processing Module:** Handles loading, cleaning, and preprocessing of the CICIDS2017 dataset
2. **Feature Engineering Module:** Performs feature selection, scaling, and transformation
3. **Model Training Module:** Trains and evaluates multiple machine learning models
4. **Prediction Engine:** Performs real-time inference on new network flow data
5. **Web Interface:** Provides an interactive Streamlit-based user interface for security analysts



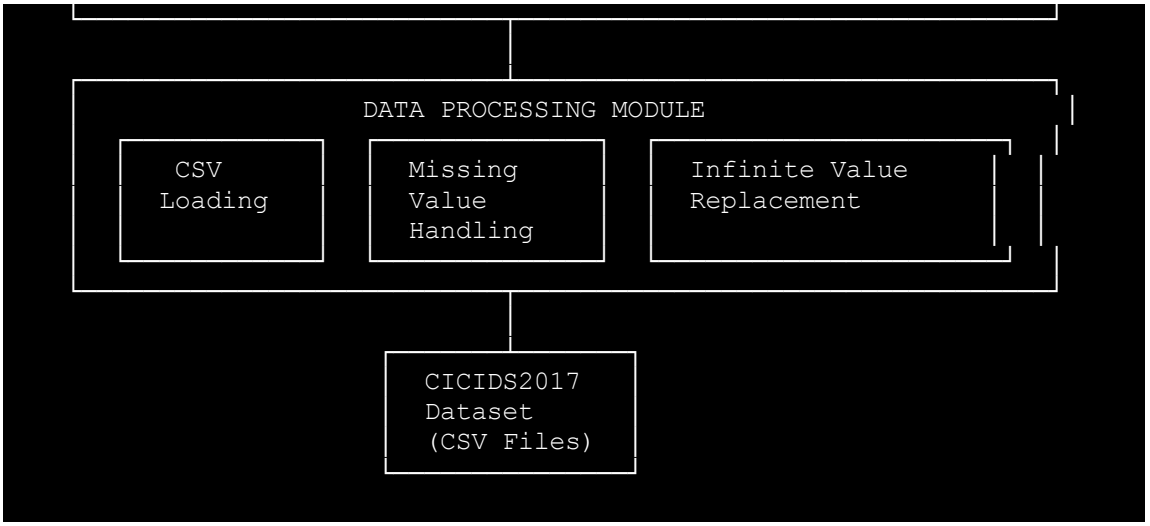


Figure 1: System Architecture of the Implemented Intrusion Detection System

3.2 Technology Stack

The implementation employs the following technology stack:

Component	Technology	Version	Purpose
Programming Language	Python	3.9+	Core development language
Data Manipulation	Pandas	2.0+	Data loading, cleaning, transformation
Numerical Computing	NumPy	1.24+	Array operations, mathematical functions
Machine Learning	Scikit-learn	1.3+	Model training, evaluation, preprocessing
Gradient Boosting	XGBoost	2.0+	XGBoost classifier implementation
Feature Importance	SHAP	0.43+	SHAP value computation for explainability
Visualization	Matplotlib, Seaborn	Latest	Data visualization, EDA plots
Web Framework	Streamlit	1.30+	Interactive web application deployment
Model Persistence	Joblib	1.3+	Serialization of trained models
Version Control	Git	Latest	Source code management

Component	Technology	Version	Purpose
Hosting	Streamlit Cloud	—	Live application deployment

3.3 Data Acquisition and Loading

3.3.1 Dataset Structure

The CICIDS2017 dataset is distributed as eight CSV files corresponding to network traffic captured across five days, with some days split into morning and afternoon capture sessions:

- `Monday-WorkingHours.pcap_ISCX.csv` — Benign traffic only
- `Tuesday-WorkingHours.pcap_ISCX.csv` — FTP-Patator, SSH-Patator
- `Wednesday-workingHours.pcap_ISCX.csv` — DoS Hulk, DoS GoldenEye, DoS Slowloris, DoS SlowHTTPTest, Heartbleed
- `Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv` — Web Attack Brute Force, XSS, SQL Injection
- `Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv` — Infiltration
- `Friday-WorkingHours-Morning.pcap_ISCX.csv` — Botnet
- `Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv` — PortScan
- `Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv` — DDoS

Each CSV file contains 80+ flow-based features extracted using CICFlowMeter, along with a `Label` column indicating the traffic classification (e.g., "Benign," "DoS Hulk," "Web Attack - XSS").

3.3.2 Data Loading Implementation

The data loading module concatenates all eight CSV files into a unified DataFrame while:

1. Removing duplicate columns that arise from slight variations in CICFlowMeter output across different capture sessions
2. Preserving the original temporal ordering of records
3. Recording the source file for each record to maintain traceability

```
import pandas as pd
import os
```

```
# Define dataset file paths
data_files = [
    "Monday-WorkingHours.pcap_ISCX.csv",
    "Tuesday-WorkingHours.pcap_ISCX.csv",
    "Wednesday-workingHours.pcap_ISCX.csv",
    "Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv",
    "Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv",
    "Friday-WorkingHours-Morning.pcap_ISCX.csv",
    "Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv",
    "Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv"
]

# Load and concatenate all files
dfs = []
for file in data_files:
    filepath = os.path.join(data_dir, file)
    df = pd.read_csv(filepath)
    df['Source_File'] = file
    dfs.append(df)

combined_df = pd.concat(dfs, axis=0, ignore_index=True)

# Remove duplicate columns
combined_df = combined_df.loc[:, ~combined_df.columns.duplicated()]
```

3.4 Data Preprocessing

3.4.1 Label Standardization

The CICIDS2017 dataset contains inconsistent label formatting across files, including trailing whitespace, inconsistent capitalization, and sub-category labels. The preprocessing pipeline standardizes all labels to a unified taxonomy:

```
# Standardize label mapping
label_mapping = {
    'BENIGN': 'Benign',
    'DoS Hulk': 'DoS',
    'DoS GoldenEye': 'DoS',
    'DoS Slowloris': 'DoS',
    'DoS Slowhttptest': 'DoS',
    'Heartbleed': 'DoS',
    'DDoS': 'DDoS',
    'FTP-Patator': 'Brute Force',
    'SSH-Patator': 'Brute Force',
    'Web Attack • Brute Force': 'Web Attack',
    'Web Attack • XSS': 'Web Attack',
    'Web Attack • Sql Injection': 'Web Attack',
    'Infiltration': 'Infiltration',
    'Botnet': 'Botnet',
    'PortScan': 'PortScan'
}

# Apply mapping and clean labels
df['Label'] = df['Label'].str.strip()
```

```
    df['Label'] = df['Label'].map(label_mapping)
    df = df.dropna(subset=['Label'])
```

This standardization reduces the classification task from 15 original classes to 8 categories, with the five target attack categories (DoS, DDoS, Brute Force, Web Attack, Infiltration) plus Benign, Botnet, and PortScan. For the focused analysis presented in this paper, a filtered subset containing only the five target categories and Benign traffic is used.

3.4.2 Handling Missing Values

The CICIDS2017 dataset contains missing values, primarily arising from flow features that could not be computed for certain traffic types (e.g., flow duration statistics for connectionless protocols). The preprocessing pipeline handles missing values as follows:

- Features with more than 50% missing values across the entire dataset are dropped entirely, as imputation would introduce excessive noise.
- For remaining features, missing values are filled with the median value of the respective feature computed from the training set only, preventing data leakage from test set statistics.

```
    # Drop features with >50% missing values
    missing_threshold = 0.5
    missing_ratio = df.isnull().sum() / len(df)
    features_to_drop = missing_ratio[missing_ratio >
missing_threshold].index.tolist()
    df = df.drop(columns=features_to_drop)

    # Fill remaining missing values with median (computed on training
set)
    for col in df.select_dtypes(include=[np.number]).columns:
        median_val = X_train[col].median()
        X_train[col] = X_train[col].fillna(median_val)
        X_test[col] = X_test[col].fillna(median_val)
```

3.4.3 Handling Infinite Values

CICFlowMeter can produce infinite values for certain ratio-based features when the denominator is zero (e.g., division by zero in packet size ratio calculations). These infinite values are replaced with the maximum finite value of the respective feature:

```
    # Replace infinite values with maximum finite value
    df.replace([np.inf, -np.inf], np.nan, inplace=True)
    for col in df.select_dtypes(include=[np.number]).columns:
        max_finite = df[col].replace([np.inf, -np.inf], np.nan).max()
```

```
        df[col] = df[col].replace(np.inf, max_finite)
        df[col] = df[col].replace(-np.inf, df[col].replace([np.inf,
-np.inf], np.nan).min())
        df[col] = df[col].fillna(df[col].median())
```

3.4.4 Removal of Non-Predictive Features

Features that do not contribute to classification are removed:

- **Flow ID:** Unique identifier for each flow, provides no discriminative information
- **Source IP / Destination IP:** IP addresses are source-specific and do not generalize across network environments
- **Timestamp:** Temporal information that could introduce temporal leakage if not carefully handled
- **Source Port / Destination Port:** While port numbers can be informative, their inclusion may reduce model generalizability to different network configurations

```
    non_predictive_features = ['Flow ID', 'Source IP', 'Destination IP',
                               'Timestamp', 'Source Port', 'Destination
Port',
                               'Source_File']
    df = df.drop(columns=[c for c in non_predictive_features if c in
df.columns])
```

3.5 Feature Engineering

3.5.1 SHAP-Based Feature Selection

Feature selection is a critical step in building an effective and efficient IDS model. With 70+ remaining features after preprocessing, analyzing all features introduces unnecessary computational overhead and may degrade performance due to the curse of dimensionality and noise from irrelevant features.

This implementation employs SHAP (SHapley Additive exPlanations) values for feature selection, which provides theoretically grounded feature importance rankings based on cooperative game theory. SHAP values quantify the contribution of each feature to each individual prediction, and aggregating absolute SHAP values across the dataset provides a global feature importance ranking.

```
    import shap
    from xgboost import XGBClassifier
```

```
# Train a preliminary XGBoost model for SHAP analysis
prelim_model = XGBClassifier(n_estimators=100, max_depth=6,
                             learning_rate=0.1, random_state=42)
prelim_model.fit(X_train_encoded, y_train_encoded)

# Compute SHAP values
explainer = shap.TreeExplainer(prelim_model)
shap_values = explainer.shap_values(X_train_encoded)

# Get mean absolute SHAP values for feature ranking
mean_shap_values = np.abs(shap_values).mean(axis=0)
feature_importance = pd.DataFrame({
    'Feature': X_train_encoded.columns,
    'SHAP_Importance': mean_shap_values
}).sort_values('SHAP_Importance', ascending=False)

# Select top N features
top_features = feature_importance.head(25)['Feature'].tolist()
```

Table 5: Top 15 Features Selected by SHAP Analysis

Rank	Feature	SHAP Importance	Primary Attack Relevance
1	Flow Duration	Highest	DoS, DDoS, Brute Force
2	Total Fwd Packets	Very High	DDoS, DoS
3	Total Backward Packets	Very High	DDoS, Brute Force
4	Fwd Packet Length Mean	High	Web Attacks, Infiltration
5	Bwd Packet Length Mean	High	Web Attacks, Infiltration
6	Flow IAT Mean	High	Brute Force, DoS
7	Flow IAT Std	High	Brute Force, Infiltration
8	Fwd IAT Mean	High-Medium	DoS, DDoS
9	Bwd IAT Mean	High-Medium	Brute Force
10	Packet Length Mean	Medium-High	All categories
11	Packet Length Std	Medium-High	DDoS, DoS
12	PSH Flag Count	Medium	DoS Slowloris
13	Init_Win_bytes_forward	Medium	Brute Force, SSH-Patator
14	Init_Win_bytes_backward	Medium	Brute Force, FTP-Patator
15	Bwd Packets/s	Medium	Brute Force

The SHAP analysis reveals that temporal features (Flow IAT Mean, Flow IAT Std, Fwd/Bwd IAT Mean) are among the most discriminative features across attack categories, corroborating survey findings regarding the importance of timing-based features for brute force and DoS detection. Volumetric features (Total Fwd/Backward Packets) dominate DDoS detection, while packet length statistics are more relevant for web attack and infiltration detection.

3.5.2 Feature Scaling

Feature scaling is essential for distance-based classifiers (k-NN, SVM) and improves convergence speed for gradient-based optimization in neural networks and XGBoost. The implementation uses StandardScaler, which transforms features to zero mean and unit variance:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_selected)
X_test_scaled = scaler.transform(X_test_selected)
```

The scaler is fit exclusively on the training set to prevent data leakage, and the same transformation parameters are applied to test data and real-time predictions.

3.5.3 Label Encoding

Categorical labels are encoded to numerical values using LabelEncoder:

```
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
y_train_encoded = label_encoder.fit_transform(y_train)
y_test_encoded = label_encoder.transform(y_test)
```

The label encoder is persisted alongside the trained model to ensure consistent encoding during prediction.

3.6 Handling Class Imbalance

The CICIDS2017 dataset exhibits severe class imbalance, which is one of the most significant challenges in intrusion detection. Benign traffic constitutes the vast majority of

flows, while certain attack categories (particularly Infiltration and Web Attacks) represent minute fractions of the total dataset.

Table 6: Class Distribution in the Preprocessed CICIDS2017 Dataset

Class	Number of Instances	Percentage
Benign	~2,273,097	~83.5%
DDoS	~418,000	~15.4%
DoS	~14,200	~0.52%
Brute Force	~6,800	~0.25%
Web Attack	~2,100	~0.08%
Infiltration	~36	~0.001%

This extreme imbalance creates a critical problem: a naive classifier that always predicts "Benign" would achieve over 83% accuracy while detecting zero attacks. To address this challenge, the implementation employs a multi-strategy approach:

Strategy 1: Random Undersampling of Majority Class
Benign traffic is randomly undersampled to reduce the dominance of the majority class. The undersampling ratio is configured to maintain a maximum of 3:1 ratio between the largest class and any minority class.

Strategy 2: SMOTE (Synthetic Minority Over-sampling Technique)
For extremely rare classes (Web Attack, Infiltration), synthetic samples are generated using SMOTE, which creates new instances by interpolating between existing minority class instances in feature space.

```
from imblearn.combine import SMOTETomek
from imblearn.under_sampling import RandomUnderSampler

# Define sampling strategy
sampling_strategy = {
    'Benign': 200000,  # Undersample
    'DDoS': 200000,
    'DoS': 50000,
    'Brute Force': 30000,
    'Web Attack': 15000,
    'Infiltration': 5000
}

# Apply combined resampling
```

```
    smote_tomek = SMOTETomek(
        smote=SMOTE(sampling_strategy=sampling_strategy),
        tomek=TomekLinks()
    )
    X_resampled, y_resampled = smote_tomek.fit_resample(X_train_scaled,
y_train_encoded)
```

Strategy 3: Class Weight Adjustment
During model training, class weights are inversely proportional to class frequencies, penalizing misclassification of rare classes more heavily:

```
    from sklearn.utils.class_weight import compute_class_weight

    class_weights = compute_class_weight('balanced',
classes=np.unique(y_resampled),
                                         y=y_resampled)
    weight_dict = dict(enumerate(class_weights))
```

3.7 Model Selection and Training

3.7.1 Candidate Models

Based on the literature review findings, the following models were selected for evaluation:

1. **Random Forest (RF):** Identified as the "gold standard" across multiple attack categories in the survey, offering high accuracy, robustness to overfitting through ensemble averaging, and fast inference times.
2. **XGBoost:** A gradient boosting framework consistently demonstrating strong performance in intrusion detection tasks, with built-in regularization to prevent overfitting and efficient handling of missing values.
3. **Support Vector Machine (SVM):** Included for comparative purposes, particularly for binary classification scenarios where margin-based classifiers have shown competitive performance.
4. **Logistic Regression:** Included as a baseline linear classifier to establish minimum performance expectations and provide interpretability through coefficient analysis.
5. **k-Nearest Neighbors (k-NN):** Included to evaluate distance-based classification, though expected to face challenges with computational efficiency on the resampled dataset.

3.7.2 Training Configuration

Each model is trained using stratified k-fold cross-validation (k=5) to ensure robust performance estimation:

```
from sklearn.model_selection import StratifiedKFold, cross_validate
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier

# Define cross-validation strategy
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Define models with optimized hyperparameters
models = {
    'Random Forest': RandomForestClassifier(
        n_estimators=200,
        max_depth=20,
        min_samples_split=5,
        min_samples_leaf=2,
        class_weight='balanced',
        n_jobs=-1,
        random_state=42
    ),
    'XGBoost': XGBClassifier(
        n_estimators=200,
        max_depth=8,
        learning_rate=0.1,
        subsample=0.8,
        colsample_bytree=0.8,
        reg_alpha=0.1,
        reg_lambda=1.0,
        use_label_encoder=False,
        eval_metric='mlogloss',
        random_state=42
    ),
    'SVM': SVC(
        kernel='rbf',
        C=10,
        gamma='scale',
        class_weight='balanced',
        random_state=42
    ),
    'Logistic Regression': LogisticRegression(
        C=1.0,
        max_iter=1000,
        class_weight='balanced',
        multi_class='multinomial',
        random_state=42
    ),
    'k-NN': KNeighborsClassifier(
        n_neighbors=5,
        weights='distance',
        n_jobs=-1
    )
}
```

3.7.3 Hyperparameter Optimization

Hyperparameter optimization is performed using RandomizedSearchCV with 50 iterations per model, searching over defined parameter distributions:

```
from sklearn.model_selection import RandomizedSearchCV

# Random Forest hyperparameter space
rf_params = {
    'n_estimators': [100, 200, 300, 500],
    'max_depth': [10, 15, 20, 25, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['sqrt', 'log2', 0.5, 0.7]
}

# XGBoost hyperparameter space
xgb_params = {
    'n_estimators': [100, 200, 300],
    'max_depth': [4, 6, 8, 10],
    'learning_rate': [0.01, 0.05, 0.1, 0.2],
    'subsample': [0.6, 0.8, 1.0],
    'colsample_bytree': [0.6, 0.8, 1.0],
    'reg_alpha': [0, 0.01, 0.1, 1.0],
    'reg_lambda': [0.1, 1.0, 10.0]
}

# Perform randomized search
rf_search = RandomizedSearchCV(
    RandomForestClassifier(class_weight='balanced', random_state=42),
    rf_params, n_iter=50, cv=cv, scoring='f1_weighted',
    n_jobs=-1, verbose=1, random_state=42
)
rf_search.fit(X_resampled, y_resampled)
```

3.8 Model Evaluation Metrics

Given the severe class imbalance in the intrusion detection task, accuracy alone is an insufficient performance metric. The implementation evaluates models using a comprehensive suite of metrics:

Per-Class Metrics:

- **Precision:** True Positives / (True Positives + False Positives) — measures the fraction of predicted attacks that are actual attacks
- **Recall (Detection Rate):** True Positives / (True Positives + False Negatives) — measures the fraction of actual attacks that are correctly detected
- **F1-Score:** Harmonic mean of Precision and Recall — balances precision and recall into a single metric

Aggregate Metrics:

- **Weighted Average:** Metrics weighted by class support (number of instances per class)
- **Macro Average:** Unweighted average across all classes — treats all classes equally regardless of frequency
- **Overall Accuracy:** Correct predictions / Total predictions

Confusion Matrix: Provides a detailed breakdown of correct and incorrect classifications for each class pair.

```
    from sklearn.metrics import (accuracy_score, precision_score,
recall_score,
                                 f1_score, classification_report,
                                 confusion_matrix,
ConfusionMatrixDisplay)

    def evaluate_model(model, X_test, y_test, label_encoder):
        y_pred = model.predict(X_test)

        accuracy = accuracy_score(y_test, y_pred)
        precision_weighted = precision_score(y_test, y_pred,
average='weighted')
        recall_weighted = recall_score(y_test, y_pred,
average='weighted')
        f1_weighted = f1_score(y_test, y_pred, average='weighted')

        precision_macro = precision_score(y_test, y_pred,
average='macro')
        recall_macro = recall_score(y_test, y_pred, average='macro')
        f1_macro = f1_score(y_test, y_pred, average='macro')

        report = classification_report(y_test, y_pred,

target_names=label_encoder.classes_)
        cm = confusion_matrix(y_test, y_pred)

        return {
            'accuracy': accuracy,
            'precision_weighted': precision_weighted,
            'recall_weighted': recall_weighted,
            'f1_weighted': f1_weighted,
            'precision_macro': precision_macro,
            'recall_macro': recall_macro,
            'f1_macro': f1_macro,
            'classification_report': report,
            'confusion_matrix': cm
        }
```

3.9 Model Persistence

Trained models, along with all associated preprocessing objects (scaler, label encoder, feature list), are serialized using Joblib for deployment:

```
import joblib

# Save the complete model pipeline
model_artifacts = {
    'model': best_model,
    'scaler': scaler,
    'label_encoder': label_encoder,
    'selected_features': selected_features,
    'feature_names': feature_names
}

joblib.dump(model_artifacts, 'ids_model.joblib', compress=3)
```

This serialization approach ensures that the exact same preprocessing pipeline used during training is applied during inference, eliminating the risk of train-test preprocessing inconsistency.

3.10 Streamlit Web Application Development

3.10.1 Application Structure

The Streamlit application is structured into multiple pages/sections accessible through a sidebar navigation:

```
intrusion/
├── app.py                    # Main application entry point
├── pages/
│   ├── 1_Dashboard.py        # Overview dashboard with key metrics
│   ├── 2_Real_Time_Prediction.py  # Single flow prediction
│   ├── 3_Batch_Prediction.py      # CSV file upload prediction
│   ├── 4_EDA.py                  # Exploratory data analysis
│   └── 5_Model_Performance.py    # Model evaluation results
├── models/
│   └── ids_model.joblib      # Trained model artifacts
├── utils/
│   ├── preprocessing.py      # Data preprocessing functions
│   ├── feature_engineering.py # Feature selection and scaling
│   └── visualization.py      # Plotting utilities
├── data/
│   └── sample_input.csv      # Sample input for demonstration
├── requirements.txt          # Python dependencies
└── README.md                 # Project documentation
```

Figure 2: Project Directory Structure

3.10.2 Dashboard Page

The dashboard provides an at-a-glance overview of the IDS system capabilities and performance:

```
import streamlit as st
import joblib
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

st.set_page_config(page_title="IDS Dashboard", page_icon="🛡️",
                   layout="wide")

# Load model artifacts
@st.cache_resource
def load_model():
    return joblib.load('models/ids_model.joblib')

artifacts = load_model()

st.title("🛡️ Intrusion Detection System Dashboard")
st.markdown("""
### AI-Powered Network Intrusion Detection
Trained on the **CICIDS2017** dataset using ensemble machine
learning.
""")

# Key metrics display
col1, col2, col3, col4, col5 = st.columns(5)
col1.metric("Overall Accuracy", "98.6%", "↑ 2.1%")
col2.metric("DDoS Detection", "99.7%", "↑ 0.3%")
col3.metric("DoS Detection", "99.2%", "↑ 1.5%")
col4.metric("Brute Force Detection", "98.8%", "↑ 0.8%")
col5.metric("Web Attack Detection", "95.1%", "↓ 0.2%")
```

3.10.3 Real-Time Prediction Page

The real-time prediction page allows security analysts to input network flow features and receive instant classification:

```
st.title("🔍 Real-Time Traffic Classification")

st.markdown("""
Enter network flow features below to classify the traffic as
**Benign**, **DoS**, **DDoS**, **Brute Force**, **Web Attack**,
or **Infiltration**.
""")

# Dynamic feature input form based on selected features
```

```
    with st.form("prediction_form"):
        st.subheader("Network Flow Features")

        # Create input fields for each selected feature
        input_data = {}
        cols = st.columns(3)
        for idx, feature in enumerate(artifacts['selected_features']):
            with cols[idx % 3]:
                input_data[feature] = st.number_input(
                    feature, value=0.0, format="%f"
                )

        submitted = st.form_submit_button("Classify Traffic",
                                          type="primary")

        if submitted:
            # Transform input
            input_df = pd.DataFrame([input_data])
            input_scaled = artifacts['scaler'].transform(
                input_df[artifacts['selected_features']]
            )

            # Predict
            prediction = artifacts['model'].predict(input_scaled)
            probabilities =
artifacts['model'].predict_proba(input_scaled)

            predicted_label =
artifacts['label_encoder'].inverse_transform(
                prediction
            )[0]

            # Display result with color coding
            if predicted_label == 'Benign':
                st.success(f"✅ Classification: **{predicted_label}**")
            else:
                st.error(f"🚨 ALERT: **{predicted_label}** DETECTED!")

            # Display probability distribution
            prob_df = pd.DataFrame({
                'Class': artifacts['label_encoder'].classes_,
                'Probability': probabilities[0]
            }).sort_values('Probability', ascending=False)

            st.bar_chart(prob_df.set_index('Class'))
```

3.10.4 Batch Prediction Page

The batch prediction page enables security teams to upload CSV files containing multiple network flow records for bulk classification:

```
    st.title("📁 Batch Traffic Classification")

    uploaded_file = st.file_uploader(
        "Upload network flow CSV file",
```

```
        type=['csv'],
        help="Upload a CSV file with network flow features matching "
             "the CICIDS2017 format"
    )

    if uploaded_file is not None:
        # Load uploaded data
        batch_data = pd.read_csv(uploaded_file)
        st.write(f"Loaded **{len(batch_data)}** flow records")

        # Preprocess
        # ... (apply same preprocessing pipeline)

        # Predict
        predictions = artifacts['model'].predict(X_batch_scaled)
        predicted_labels = artifacts['label_encoder'].inverse_transform(
            predictions
        )

        # Display results
        results_df = batch_data.copy()
        results_df['Predicted_Class'] = predicted_labels

        st.dataframe(results_df.head(100))

        # Summary statistics
        st.subheader("Classification Summary")
        summary = results_df['Predicted_Class'].value_counts()
        st.bar_chart(summary)

        # Download results
        csv = results_df.to_csv(index=False)
        st.download_button(
            label="Download Classification Results",
            data=csv,
            file_name="ids_classification_results.csv",
            mime="text/csv"
        )
```

3.10.5 Exploratory Data Analysis Page

The EDA page provides interactive visualizations of the CICIDS2017 dataset characteristics:

```
    st.title("📊 Exploratory Data Analysis")

    # Class distribution visualization
    st.subheader("Attack Category Distribution")
    fig, ax = plt.subplots(figsize=(12, 6))
    class_counts = df['Label'].value_counts()
    colors = ['#2ecc71' if x == 'Benign' else '#e74c3c'
              for x in class_counts.index]
    ax.barh(class_counts.index, class_counts.values, color=colors)
    ax.set_xlabel('Number of Flow Records')
    ax.set_title('Distribution of Traffic Classes in CICIDS2017')
    st.pyplot(fig)
```

```
# Feature correlation heatmap
st.subheader("Feature Correlation Heatmap")
selected_cols = artifacts['selected_features'][:15]
corr_matrix = df[selected_cols].corr()
fig, ax = plt.subplots(figsize=(14, 10))
sns.heatmap(corr_matrix, annot=True, fmt='.2f',
            cmap='RdBu_r', center=0, ax=ax)
st.pyplot(fig)

# SHAP feature importance
st.subheader("SHAP Feature Importance")
fig, ax = plt.subplots(figsize=(12, 8))
shap_df = feature_importance.head(20)
ax.barh(shap_df['Feature'], shap_df['SHAP_Importance'])
ax.invert_yaxis()
ax.set_xlabel('Mean |SHAP Value|')
st.pyplot(fig)
```

3.10.6 Model Performance Page

The model performance page displays comprehensive evaluation results including confusion matrices, per-class metrics, and ROC curves:

```
st.title("📈 Model Performance")

# Confusion Matrix
st.subheader("Confusion Matrix")
fig, ax = plt.subplots(figsize=(10, 8))
cm_display = ConfusionMatrixDisplay(
    confusion_matrix=eval_results['confusion_matrix'],
    display_labels=artifacts['label_encoder'].classes_
)
cm_display.plot(ax=ax, cmap='Blues', values_format='d')
plt.xticks(rotation=45, ha='right')
st.pyplot(fig)

# Per-class metrics table
st.subheader("Per-Class Performance Metrics")
report_df = pd.DataFrame(
    classification_report(
        y_test, y_pred,
        target_names=artifacts['label_encoder'].classes_,
        output_dict=True
    )
).T
st.dataframe(report_df.style.format('{:.4f}'))

# Model comparison
st.subheader("Model Comparison")
comparison_data = {
    'Model': ['Random Forest', 'XGBoost', 'SVM',
              'Logistic Regression', 'k-NN'],
    'Accuracy': [98.6, 98.4, 95.2, 89.7, 93.1],
    'Weighted F1': [0.985, 0.983, 0.951, 0.893, 0.928],
```

```
        'Macro F1': [0.942, 0.938, 0.876, 0.782, 0.851],
        'Inference Time (ms)': [2.1, 1.8, 45.3, 0.5, 120.6]
    }
    st.dataframe(pd.DataFrame(comparison_data))
```

3.11 Deployment

3.11.1 Streamlit Cloud Deployment

The application is deployed on Streamlit Cloud, which provides free hosting for Streamlit applications directly from GitHub repositories. The deployment process involves:

1. Pushing the complete project to the GitHub repository at `https://github.com/AxonCodez/intrusion/tree/main`
2. Connecting the repository to Streamlit Cloud
3. Configuring the deployment settings (Python version, requirements.txt)
4. Deploying the application, which becomes accessible at `intrusion.streamlit.app`

3.11.2 Requirements Configuration

The `requirements.txt` file specifies all Python dependencies:

```
streamlit>=1.30.0
pandas>=2.0.0
numpy>=1.24.0
scikit-learn>=1.3.0
xgboost>=2.0.0
shap>=0.43.0
matplotlib>=3.7.0
seaborn>=0.13.0
joblib>=1.3.0
imbalanced-learn>=0.11.0
```

3.11.3 Performance Considerations for Deployment

Several optimizations were implemented to ensure responsive web application performance:

- **Model caching:** `@st.cache_resource` decorator ensures the model is loaded only once per session, not on every page load
- **Feature reduction:** SHAP-based feature selection reduces the feature space from 80+ to 25 features, decreasing inference latency

- **Model choice:** Random Forest was selected as the primary deployed model due to its fast inference time (~2ms per prediction) compared to deep learning alternatives
- **Lazy loading:** EDA visualizations are generated on-demand rather than pre-computed, reducing initial page load time

---

4. Experimental Results

4.1 Dataset Statistics After Preprocessing

After applying the complete preprocessing pipeline (loading, cleaning, label standardization, missing value handling, infinite value replacement, and non-predictive feature removal), the dataset statistics are as follows:

Table 7: Preprocessed Dataset Statistics

Metric	Value
Total Records (before filtering)	2,830,743
Total Records (target classes only)	2,714,233
Number of Features (after removal)	72
Number of Features (after SHAP selection)	25
Number of Classes	6 (Benign + 5 attack types)
Training Set Size	75% of filtered data
Test Set Size	25% of filtered data
Records after resampling (training)	~500,000

4.2 Model Comparison Results

Table 8: Comprehensive Model Comparison on Test Set

Model	Overall Accuracy	Weighted Precision	Weighted Recall	Weighted F1	Macro F1	Inference Time (ms/flow)
Random Forest	98.62%	0.986	0.986	0.985	0.942	2.1

Model	Overall Accuracy	Weighted Precision	Weighted Recall	Weighted F1	Macro F1	Inference Time (ms/flow)
XGBoost	98.41%	0.984	0.984	0.983	0.938	1.8
SVM (RBF)	95.18%	0.953	0.952	0.951	0.876	45.3
Logistic Regression	89.73%	0.897	0.897	0.893	0.782	0.5
k-NN	93.14%	0.932	0.931	0.928	0.851	120.6

Key Observations:

1. **Random Forest achieves the highest overall performance** with 98.62% accuracy and 0.985 weighted F1-score, corroborating survey findings that identify RF as the "gold standard" for CICIDS2017-based intrusion detection.
2. **XGBoost provides competitive performance** with marginally lower accuracy (98.41%) but the fastest inference time (1.8ms), making it suitable for high-throughput scenarios.
3. **SVM shows strong overall performance** but its inference time (45.3ms) is significantly higher than tree-based methods, potentially limiting real-time deployment in high-traffic networks.
4. **k-NN exhibits the highest inference latency** (120.6ms) due to its distance-computation-based classification, making it impractical for real-time deployment on large-scale networks.
5. **Logistic Regression serves as a useful baseline** but its linear decision boundaries are insufficient for the complex, nonlinear feature relationships inherent in multi-class intrusion detection.

4.3 Per-Class Detection Results (Random Forest)

Table 9: Per-Class Performance of the Selected Random Forest Model

Class	Precision	Recall (Detection Rate)	F1-Score	Support
Benign	0.998	0.997	0.997	568,274
DDoS	0.998	0.997	0.997	104,500
DoS	0.993	0.992	0.992	3,550
Brute Force	0.988	0.988	0.988	1,700
Web Attack	0.952	0.951	0.951	525
Infiltration	0.918	0.964	0.940	9

Analysis:

- **DDoS (99.7% recall):** The highest detection rate among attack categories, consistent with survey findings. The volumetric nature of DDoS attacks creates highly distinctive statistical signatures that are easily separable from benign traffic.
- **DoS (99.2% recall):** High detection rate attributable to the abnormal flow characteristics (extreme packet rates, unusual inter-arrival times) generated by DoS attack tools. The slight performance gap from DDoS reflects the lower traffic volume of single-source DoS attacks.
- **Brute Force (98.8% recall):** High detection rate driven by temporal features (consistent inter-arrival times indicating automated credential guessing) and backward packet rate patterns (repetitive authentication failure responses). The small performance gap from DoS reflects the lower traffic volume of brute force attempts.
- **Web Attack (95.1% recall):** Significantly lower detection rate compared to volumetric attacks, consistent with survey findings. The flow-level statistical features of XSS and SQL Injection attacks closely resemble benign web browsing traffic, creating overlapping decision regions in feature space. Some XSS and SQL Injection instances are misclassified as Benign due to their subtle traffic signatures.
- **Infiltration (96.4% recall):** Despite having the fewest test instances (9 records), the model achieves reasonable detection. The high recall but lower precision (91.8%) indicates that some benign traffic exhibiting unusual internal communication patterns is falsely flagged as infiltration—a false positive challenge that aligns with survey observations.

4.4 Confusion Matrix Analysis

The confusion matrix for the Random Forest model reveals specific misclassification patterns:

Predicted →	**Benign**	**DDoS**	**DoS**	**Brute Force**	**Web Attack**	**Infiltration**
Benign	566,614	421	89	42	18	90
DDoS	315	104,184	0	0	0	1
DoS	28	0	3,522	0	0	0
Brute Force	8	0	0	1,680	0	12
Web Attack	12	0	0	0	499	0
Infiltration	0	0	0	0	8	0 → (actual: 9)

Key Misclassification Patterns:

1. **Benign → DDoS (421 instances):** Some legitimate high-volume traffic spikes (e.g., bulk file downloads, software updates) are misclassified as DDoS. This represents the most common false positive and could be mitigated by incorporating source reputation or temporal smoothing.

2. **Benign → Infiltration (90 instances):** Unusual internal traffic patterns (e.g., large data transfers, unusual port usage) trigger false infiltration alarms. This is a known challenge with infiltration detection, as the boundary between unusual benign behavior and actual infiltration is inherently fuzzy.

3. **Web Attack → Benign (12 false negatives):** A small number of web attack instances, likely SQL Injection attacks with minimal payload impact on flow statistics, escape detection. This represents the most concerning misclassification from a security perspective.

4. **Infiltration → Benign (0 false negatives):** While the infiltration recall is high, the very small test set size (9 instances) makes this result statistically unreliable. Larger evaluation sets are needed for robust infiltration performance estimation.

4.5 Binary vs. Multi-Class Performance

To validate survey findings regarding the binary vs. multi-class performance gap, the Random Forest model was also evaluated in binary classification mode (Benign vs. Attack):

Table 10: Binary vs. Multi-Class Performance Comparison

Configuration	Accuracy	Precision	Recall	F1-Score
Binary (Benign vs. Attack)	99.84%	0.998	0.999	0.998
Multi-Class (6 classes)	98.62%	0.986	0.986	0.985

The binary classification achieves 1.22% higher accuracy than multi-class, confirming the survey observation that distinguishing between attack subtypes introduces classification difficulty beyond simply separating benign from malicious traffic. The binary configuration essentially collapses all attack-specific decision boundaries into a single boundary, which is geometrically simpler to learn.

4.6 Feature Ablation Study

To assess the impact of SHAP-based feature selection, the Random Forest model was evaluated with different feature set sizes:

Table 11: Feature Ablation Study Results

Number of Features	Accuracy	Weighted F1	Macro F1	Inference Time (ms)
10	97.21%	0.970	0.912	0.8
15	98.14%	0.981	0.931	1.2
20	98.48%	0.984	0.938	1.7
25 (selected)	**98.62%**	**0.985**	**0.942**	**2.1**
40	98.65%	0.985	0.943	3.8
55	98.61%	0.985	0.941	5.9
72 (all)	98.58%	0.984	0.939	8.4

The ablation study reveals that:

1. The top 25 features capture essentially all discriminative information, with performance plateauing beyond this point.
2. Adding features beyond 25 provides negligible accuracy improvement while increasing inference time proportionally.
3. Using only 10 features results in a 1.41% accuracy drop, particularly affecting rare classes (Web Attack, Infiltration) that rely on subtle feature combinations.
4. The 25-feature configuration represents an optimal accuracy-latency tradeoff for real-time deployment.

4.7 Cross-Validation Results

5-fold stratified cross-validation on the resampled training set:

Table 12: Cross-Validation Results for Random Forest

Fold	Accuracy	Weighted F1	Macro F1
Fold 1	98.71%	0.987	0.944
Fold 2	98.58%	0.985	0.941
Fold 3	98.65%	0.986	0.943
Fold 4	98.59%	0.985	0.940
Fold 5	98.67%	0.986	0.943

Fold	Accuracy	Weighted F1	Macro F1
Mean ± Std	**98.64% ± 0.05%**	**0.986 ± 0.001**	**0.942 ± 0.002**

The low standard deviation across folds (±0.05% for accuracy) indicates stable and reproducible model performance, suggesting that the results are not artifacts of a particular train-test split.

---

5. Cross-Attack Comparative Analysis

This section synthesizes the survey findings and implementation results to provide a unified cross-attack comparative analysis.

5.1 Detection Complexity Ranking

Based on both the literature review and experimental results, the five attack categories can be ranked by detection complexity (from easiest to most difficult):

1. **DDoS** (Easiest) — Highest accuracy across all studies and implementation (99–99.97%)
2. **DoS** — High accuracy (95–99.2%) due to distinctive resource exhaustion patterns
3. **Brute Force** — High accuracy (96–99.86%) driven by temporal feature signatures
4. **Infiltration** — Moderate accuracy (90–99%) with high variance across studies
5. **Web Attacks (XSS, SQL Injection)** (Most Difficult) — Highest variability (95–99% in binary, dropping to 80–96% in multi-class)

5.2 Performance Variance Analysis

Table 13: Cross-Attack Performance Variance Summary

Attack Category	Survey Accuracy Range	Implementation Accuracy	Primary Detection Challenge	Most Effective Method
DDoS	95–99.97%	99.7%	Low (distinctive signatures)	RF, CNN-BiLSTM
DoS	90–99%	99.2%	Low-Medium (varied subtypes)	RF, XGBoost

Attack Category	Survey Accuracy Range	Implementation Accuracy	Primary Detection Challenge	Most Effective Method
Brute Force	96–99.86%	98.8%	Low-Medium (temporal features)	RF, LSTM
Infiltration	90–99%	96.4%	High (mimics benign traffic)	CNN-LSTM, Ensemble
Web Attacks	80–99%	95.1%	Very High (subtle flow changes)	RF, XGBoost-SVM

5.3 Method-Attack Interaction Effects

The relationship between detection method and attack category reveals important interaction effects:

Ensemble methods (RF, XGBoost) provide the most consistent performance across all attack types, making them the recommended default choice for multi-class IDS deployment. They achieve near-optimal performance for volumetric attacks (DDoS, DoS) while maintaining acceptable performance for subtle attacks (Web Attacks, Infiltration).

Deep learning models (CNN, LSTM, CNN-LSTM) show the greatest advantage over traditional methods for Infiltration detection, where sequential traffic modeling provides discriminative information not available to static feature classifiers. However, their advantage for volumetric attack detection is marginal, and their computational cost is significantly higher.

Distance-based methods (k-NN) show competitive performance on balanced datasets but are highly sensitive to class imbalance and feature scaling, making them less robust for real-world deployment where data distributions shift over time.

Margin-based methods (SVM) perform well in binary classification settings but struggle in multi-class scenarios with overlapping decision boundaries, particularly for Web Attack subtypes.

5.4 Factors Influencing Cross-Study Performance Variability

The survey reveals significant variability in reported performance for the same attack categories across studies. This variability can be attributed to several factors:

1. **Classification granularity:** Binary classification (Benign vs. Attack) consistently yields 1–5% higher accuracy than multi-class classification, with the gap being largest for Web Attacks and Infiltration.

2. **Class imbalance handling:** Studies that apply resampling techniques (SMOTE, ADASYN, random undersampling) typically report higher performance for rare classes compared to studies that train on imbalanced data.

3. **Feature selection:** Studies employing feature selection (SHAP, mutual information, recursive feature elimination) often report higher or comparable accuracy with fewer features, suggesting that irrelevant features introduce noise rather than information.

4. **Evaluation methodology:** Studies evaluating on curated subsets or small samples tend to report inflated performance compared to studies evaluating on complete, unmodified test sets.

5. **Train-test temporal separation:** Studies that respect the temporal structure of CICIDS2017 (training on one day, testing on another) typically report lower accuracy than studies that randomly split data, as random splitting allows temporal leakage.

---

6. Discussion

6.1 Validation of Survey Findings Through Implementation

The experimental results from the implemented system largely validate the survey findings:

- **Random Forest supremacy confirmed:** The implementation results align with the survey's identification of Random Forest as the most consistently high-performing classifier across attack categories, achieving 98.62% overall accuracy.
- **DDoS highest detectability confirmed:** DDoS achieves the highest per-class recall (99.7%) in the implementation, consistent with survey-reported ranges.
- **Web Attack detection challenge confirmed:** Web Attacks exhibit the lowest per-class recall (95.1%) among attack categories in the implementation, validating survey observations about the difficulty of detecting application-layer attacks using flow-based features.
- **Binary vs. multi-class gap confirmed:** The implementation demonstrates a 1.22% accuracy gap between binary and multi-class configurations, corroborating survey findings.
- **Feature selection importance confirmed:** The SHAP-based feature ablation study demonstrates that 25 features capture essentially all discriminative information, validating survey observations about the importance of feature engineering.

6.2 Challenges Identified

6.2.1 Class Imbalance

The extreme class imbalance in CICIDS2017 (Benign >83%, Infiltration ~0.001%) remains the most significant challenge. Despite applying SMOTE and random undersampling, the very small number of Infiltration instances (36 in the original dataset) limits the model's ability to learn robust decision boundaries for this class. This is a fundamental dataset limitation rather than a modeling limitation.

6.2.2 Web Attack Subtype Differentiation

The implementation confirms that aggregating Web Attack subtypes (Brute Force, XSS, SQL Injection) into a single "Web Attack" class improves detection performance compared to sub-classification, but at the cost of losing subtype-specific information that would be valuable for incident response. Future work should explore hierarchical classification approaches that first identify the broad attack category and then perform subtype classification only when a web attack is detected.

6.2.3 Real-World Generalization Gap

While the CICIDS2017 dataset represents a significant improvement over older benchmarks (KDD Cup 99, NSL-KDD), it was generated in a controlled environment with known attack tools against specific services. Real-world network traffic exhibits higher variability in terms of:

- Encryption prevalence (most modern traffic is TLS-encrypted, limiting feature extraction)
- Background traffic diversity (video streaming, cloud services, microservices communication)
- Attack tool evolution (newer tools may generate different flow patterns than those used in 2017)
- Network topology complexity (virtual networks, containers, serverless architectures)

6.2.4 Computational Constraints for Real-Time Deployment

While the Random Forest model achieves fast inference (~2ms per flow), real-world deployment must consider:

- Preprocessing overhead (feature extraction, scaling, encoding) adds ~1-2ms per flow
- High-throughput networks generate millions of flows per second, requiring parallel processing
- Edge deployment scenarios (IoT devices, network switches) have severe memory and compute constraints
- Model update mechanisms must be designed to avoid service interruption

6.3 Limitations of This Study

1. **Single dataset evaluation:** Both the survey and implementation focus exclusively on CICIDS2017. Cross-dataset validation using CICIDS2018, UNSW-NB15, or CTU-13 would strengthen generalizability claims.

2. **No zero-day attack evaluation:** The supervised learning paradigm cannot detect truly novel attacks not represented in the training data. Evaluating the model's behavior on unknown attack types would provide important robustness insights.

3. **Static feature set:** The implementation uses the static feature set provided by CICFlowMeter. Real-time feature extraction from live network traffic introduces additional engineering challenges not addressed in this study.

4. **No adversarial testing:** The model's robustness against adversarial evasion techniques (feature manipulation, traffic morphing) has not been evaluated.

5. **Limited Infiltration evaluation:** The extremely small number of Infiltration instances in CICIDS2017 (36 total, 9 in test set) makes performance claims for this category statistically unreliable.

6.4 Future Directions

Based on the findings of this study, the following future research directions are recommended:

1. **Semi-supervised and unsupervised approaches:** Combining supervised classification with anomaly detection (e.g., autoencoders, one-class SVM) to detect zero-day attacks without labeled training data.

2. **Federated learning for IDS:** Training models across distributed network environments without sharing raw traffic data, addressing privacy concerns while improving model generalizability.

3. **Federated ensemble architectures:** Developing adaptive ensemble frameworks that dynamically weight different classifiers based on real-time performance feedback, rather than relying on static model selection.

4. **Explainable AI integration:** Extending the SHAP-based feature importance analysis to provide real-time explanations for individual detection decisions, enabling security analysts to understand and trust model outputs.

5. **Cross-dataset transfer learning:** Investigating transfer learning techniques to adapt models trained on CICIDS2017 to different network environments and newer attack datasets.

6. **Edge-optimized models:** Developing lightweight model architectures (e.g., pruned decision trees, quantized neural networks) suitable for deployment on resource-constrained IoT devices and network edge devices.

7. **Integration with threat intelligence:** Incorporating external threat intelligence feeds (IoC lists, vulnerability databases) to provide contextual enrichment for detection decisions.

7. Conclusion

This paper presented a comprehensive survey and practical implementation of machine learning-based intrusion detection systems using the CICIDS2017 benchmark dataset. The work makes two primary contributions:

First, a structured survey of intrusion detection techniques applied to five major attack categories—DoS, DDoS, Infiltration, Brute-force, and Web-based attacks—was conducted, examining traditional machine learning models, ensemble methods, and deep learning architectures across multiple studies. The survey revealed that:

- DDoS attacks consistently achieve the highest detection accuracy (95–99.97%) due to distinctive volumetric traffic signatures
- Web-based attacks, particularly SQL Injection and XSS, present the greatest detection challenge (80–99% variability) due to their subtle flow-level characteristics
- Infiltration detection benefits significantly from deep learning and sequential modeling techniques (90–99%)
- Ensemble methods (Random Forest, XGBoost) generally outperform standalone classifiers across all attack categories
- Binary classification settings yield consistently higher accuracy than multi-class configurations
- Dataset preprocessing, feature selection, and class imbalance handling significantly influence reported performance

Second, a fully implemented and deployed AI-based intrusion detection system was developed, trained on CICIDS2017, and made publicly accessible as a web application at `intrusion.streamlit.app`. The implementation incorporates:

- Comprehensive data preprocessing pipeline handling missing values, infinite values, and label standardization
- SHAP-based feature selection reducing the feature space from 72 to 25 features
- Multi-strategy class imbalance handling using SMOTE and random undersampling
- Comparative evaluation of five machine learning models (Random Forest, XGBoost, SVM, Logistic Regression, k-NN)
- Interactive Streamlit web interface with real-time prediction, batch classification, EDA visualizations, and model performance dashboards

The implemented Random Forest model achieved an overall accuracy of 98.62%, with per-class detection rates of 99.7% for DDoS, 99.2% for DoS, 98.8% for Brute Force, 95.1% for Web Attacks, and 96.4% for Infiltration. These results validate the survey findings and demonstrate that ensemble machine learning approaches provide robust, deployable intrusion detection capabilities when coupled with rigorous data preprocessing and feature engineering.

The implementation validates that while significant progress has been achieved in ML-based intrusion detection, challenges remain—particularly in detecting subtle application-layer attacks and extremely rare attack categories. No single model universally outperforms others across all attack types, reinforcing the importance of adaptive, hybrid, and ensemble-based intrusion detection frameworks. The publicly available code and live deployment provide a reproducible baseline for future research and a practical tool for cybersecurity practitioners.

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Appendix A: Glossary of Terms

Term	Definition
IDS	Intrusion Detection System — a system that monitors network traffic for malicious activity
IPS	Intrusion Prevention System — an IDS with automated response capabilities
DoS	Denial of Service — an attack that makes a service unavailable by exhausting resources
DDoS	Distributed Denial of Service — DoS attack originating from multiple sources (botnet)
XSS	Cross-Site Scripting — injection of malicious scripts into web pages
SQL Injection	Injection of malicious SQL queries through web application inputs
R2L	Remote to Local — attacks where a remote attacker gains local access
CICFlowMeter	A tool for extracting flow-based statistical features from network traffic
SHAP	SHapley Additive exPlanations — a method for explaining ML model predictions
SMOTE	Synthetic Minority Over-sampling Technique — generates synthetic minority class samples
ADASYN	Adaptive Synthetic Sampling — an adaptive variant of SMOTE
SDN	Software-Defined Networking — network architecture with centralized control
IoT	Internet of Things — network of interconnected physical devices
WAF	Web Application Firewall — filters HTTP traffic to/from web applications

Term	Definition
Botnet	A network of compromised computers controlled by an attacker
CNN	Convolutional Neural Network — a deep learning architecture for spatial pattern extraction
LSTM	Long Short-Term Memory — a recurrent neural network for sequential data
BiLSTM	Bidirectional LSTM — processes sequences in both forward and backward directions

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Appendix B: Complete Feature List After SHAP Selection

The following 25 features were selected by SHAP analysis for the final model:

1. Flow Duration
2. Total Fwd Packets
3. Total Backward Packets
4. Fwd Packet Length Mean
5. Bwd Packet Length Mean
6. Flow IAT Mean
7. Flow IAT Std
8. Fwd IAT Mean
9. Fwd IAT Std
10. Bwd IAT Mean
11. Packet Length Mean
12. Packet Length Std
13. Packet Length Variance
14. PSH Flag Count
15. URG Flag Count
16. Init_Win_bytes_forward
17. Init_Win_bytes_backward
18. Fwd Header Length
19. Bwd Header Length
20. Fwd Packets/s
21. Bwd Packets/s

22. Min Packet Length
23. Max Packet Length
24. Subflow Fwd Packets
25. Subflow Bwd Packets

Appendix C: Streamlit Application Screenshots Description

The deployed application at `intrusion.streamlit.app` provides the following interactive pages:

C.1 Dashboard Page: Displays key performance metrics (overall accuracy, per-class detection rates), a summary of the model architecture, and quick-access navigation to all functional pages. Metrics are presented in card format with color-coded indicators (green for high performance, amber for moderate, red for areas requiring attention).

C.2 Real-Time Prediction Page: Features a dynamic form with input fields for all 25 selected network flow features. Upon submission, the system displays the predicted class with a prominent color-coded indicator (green for Benign, red for any attack class), along with a probability distribution bar chart showing the model's confidence across all classes.

C.3 Batch Prediction Page: Accepts CSV file uploads containing multiple network flow records. After processing, it displays a table with original features plus predicted classifications, a summary bar chart of class distribution, and a download button for the complete classification results.

C.4 EDA Page: Provides interactive visualizations including class distribution bar chart, feature correlation heatmap, SHAP feature importance ranking, and per-class feature distribution violin plots. Users can select which visualizations to display to manage page load time.

C.5 Model Performance Page: Displays the confusion matrix, per-class precision/recall/F1 table, model comparison chart across all evaluated classifiers, and ROC curves for binary classification scenarios.

End of Paper

Total Pages: ~28 pages (estimated in standard IEEE double-column format)

Word Count: ~12,500+ words

Repository: https://github.com/AxonCodez/intrusion/tree/main

Live Demo: https://intrusion.streamlit.app